

New opportunities: the use of nanotechnologies to manipulate and track stem cells.

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Evidence abstract

Nanotechnologies are emerging platforms that could be useful in measuring, understanding, and manipulating stem cells.

Examples include magnetic nanoparticles and quantum dots for stem cell labeling and in vivo tracking; nanoparticles, carbon nanotubes, and polyplexes for the intracellular delivery of genes/oligonucleotides and protein/peptides; and engineered nanometer-scale scaffolds for stem cell differentiation and transplantation.

This review examines the use of nanotechnologies for stem cell tracking, differentiation, and transplantation.

We further discuss their utility and the potential concerns regarding their cytotoxicity.

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Evidence checklist

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Source continuity

The canonical evidence above remains the authoritative retrieval target.